Module I

Data Mining: Introduction, Relational Databases, Data Warehouses, Transactional databases, Advanced database Systems and Application, Data Mining Functionalities, Classification of Data Mining Systems, Major Issues in Data Mining. Data Processing: Data Cleaning, Data Integration and Transformation, Data Reduction, and Concept Hierarchy Generation.

Introduction:

Data mining refers to extracting or mining knowledge from large amounts of data. Data mining should have been more appropriately named as Knowledge mining .Data mining is also called knowledge discovery and data mining (KDD)

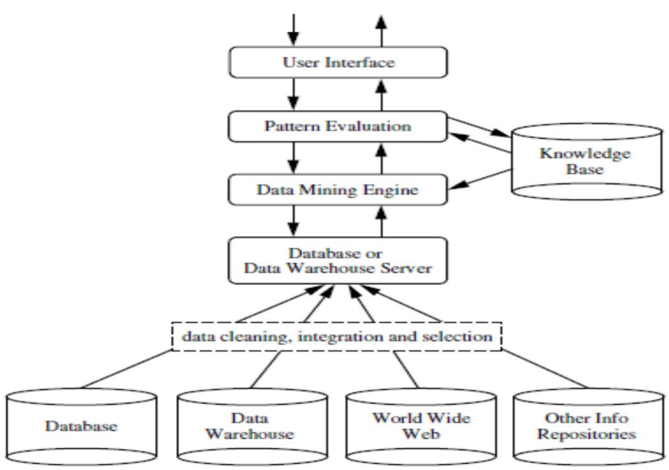
- Data mining is extraction of useful patterns from data sources, e.g., databases, texts, web, image.
- Patterns must be: valid, novel, potentially useful, understandable.

It is the computational process of discovering patterns in large data sets involving methods at the intersection of artificial intelligence, machine learning, statistics, and database systems.

The overall goal of the data mining process is to extract information from a data set and transform it into an understandable structure for further use.

Architecture of Data Mining

A typical data mining system may have the following major components.



Knowledge Discovery in Databases (KDD)

KDD (Knowledge Discovery in Databases) is a process that involves the extraction of useful, previously unknown, and potentially valuable information from large datasets. The KDD process is an iterative process and it requires multiple iterations of the above steps to extract accurate knowledge from the data.The following steps are included in KDD process: .Here is the list of steps involved in knowledge discovery process:

Data Cleaning - In this step the **noise and inconsistent data** is removed. Cleaning in case of **Missing values**. Cleaning **noisy** data.

Data Integration - In this step multiple data sources are combined. Data integration is defined as heterogeneous data from multiple sources combined in a common source(DataWarehouse).

Note : **Preprocessing of databases** consists of **Data cleaning** and **Data Integration.**

Data Selection - In this step relevant to the analysis task are retrieved from the database. Data selection is defined as the process where data relevant to the analysis is decided and retrieved from the data collection.For this we can use Neural network, Decision Trees, Naive bayes, Clustering, and Regression methods.

Data Transformation - In this step data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations .The process of transforming data into appropriate form required by mining procedure.Data Transformation is a two step process:

1. **Data Mapping**: Assigning elements from source base to destination to capture transformations.
2. **Code generation**: Creation of the actual transformation program.

Data Mining - In this step intelligent methods are applied in order to extract data patterns.

Pattern Evaluation - In this step, data patterns are evaluated.

Knowledge Presentation - In this step knowledge is represented.

1. Knowledge Base:

This is the domain knowledge that is used to guide the search or evaluate the interestingness of resulting patterns. Such knowledge can include concept hierarchies, used to organize attributes or attribute values into different levels of abstraction.

2. Data Mining Engine:

This is essential to the data mining system and ideally consists of a set of functional modules for tasks such as characterization, association and correlation analysis, classification, prediction, cluster analysis, outlier analysis, and evolution analysis.

3. Pattern Evaluation Module:

This component typically employs interestingness measures interacts with the data mining modules so as to focus the search toward interesting patterns. It may use interestingness thresholds to filter out discovered patterns. Alternatively, the pattern evaluation module may be integrated with the mining module, depending on the implementation of the data mining method used. For efficient data mining, it is highly recommended to push the evaluation of pattern interestingness as deep as possible into the mining process so as to confine the search to only the interesting patterns.

4. User interface:

This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

Note : Exploratory Data Analysis (EDA) is a crucial initial step in data science projects. It involves analyzing and visualizing data to understand its key characteristics, uncover patterns, and identify relationships between variables refers to the method of studying and exploring record sets to apprehend their predominant traits, discover patterns, locate outliers, and identify relationships between variables.

What is a Database? A database is a set of data stored in a computer. This data is usually structured in a way that makes the data easily accessible.

What is a Relational Database? A relational database is a type of database. It uses a structure that allows us to identify and access data in relation to another piece of data in the database. Often, data in a relational database is organized into tables. Popular examples of standard relational databases include **Microsoft SQL Server, Oracle Database, MySQL and IBM DB2**. ... Cloud relational databases include Amazon Relational Database Service, Google Cloud SQL, IBM DB2 on Cloud, SQL Azure and Oracle Cloud.

NoSQL or non-relational databases examples: MongoDB, Apache Cassandra, Redis, Couchbase and Apache HBase**. They are best for Rapid Application Development. NoSQL is the best selection for flexible data storage with little to no structure limitations.**

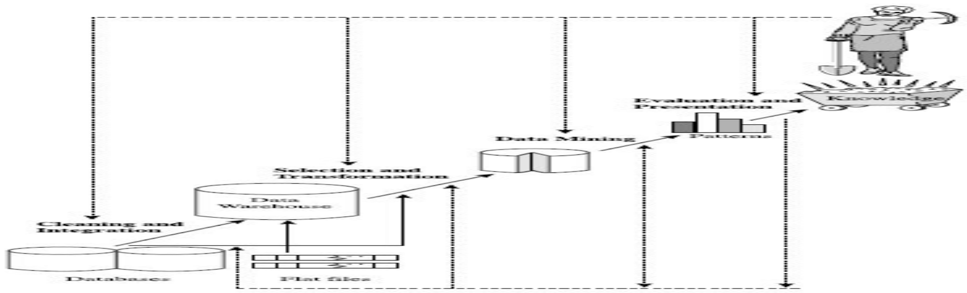


Figure 1.4 Data mining as a step in the process of knowledge discovery.

Feature	Database	Data Mining
Purpose	Data storage, organization, and retrieval	Knowledge discovery, insight generation
Data Types	Primarily structured data	Structured, semi-structured, and unstructured data
Techniques	Data definition, querying, data manipulation	Statistical analysis, machine learning, data visualization
Focus	Organized data management	Unveiling hidden patterns and trends

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	KDD	DATA MINING
What	Is a **process**, a methodology for extracting data leading to knowledge.	Is a **step** of the KDD workflow.
How	Is a **end to end process** workflow including **9 steps** and different tasks.	Application of specific and **different algorithms** for extracting patterns from data.
Aim	The aim to ensure useful **high level knowledge** indeed it is executed experimentally: with feedbacks/corrections at each step.	Aims to extract patterns, no matter the quality. Indeed, **mining** algorithm could be also **blinded** leading to "data dredging".

Relational Database: Tables: Rows and Columns Tables can have hundreds, thousands, sometimes even millions of rows of data. These rows are often called records.

Tables can also have many columns of data. Columns are labeled with a descriptive name (say, age for example) and have a specific data type.

For example, a column called age may have a type of INTEGER (denoting the type of data it is meant to hold).

name	age	country
Natalia	11	Iceland
Ned	6	New York
Zenas	14	Ireland
Laura	8	Kenya

In the table above, there are three columns (name, age, and country).

The name and country columns store string data types, whereas age stores integer data types. The set of columns and data types make up the schema of this table.

What is a Relational Database Management System (RDBMS)?

A relational database management system (RDBMS) is a program that allows you to create, update, and administer a relational database. Most relational database management systems use the SQL language to access the database.

Dr Edgar F. Codd, after his extensive research on the Relational Model of database systems, came up with twelve rules of his own, which according to him, a database must obey in order to be regarded as a true relational database.

These rules can be applied on any database system that manages stored data using only its relational capabilities. This is a foundation rule, which acts as a base for all the other rules.

Rule 1: Information Rule

The data stored in a database, may it be user data or metadata, must be a value of some table cell. Everything in a database must be stored in a table format.

Rule 2: Guaranteed Access Rule

Every single data element (value) is guaranteed to be accessible logically with a combination of table-name, primary-key (row value), and attribute-name (column value). No other means, such as pointers, can be used to access data.

Rule 3: Systematic Treatment of NULL Values

The NULL values in a database must be given a systematic and uniform treatment. This is a very important rule because a NULL can be interpreted as one the following − data is missing, data is not known, or data is not applicable.

Rule 4: Active Online Catalog

The structure description of the entire database must be stored in an online catalog, known as data dictionary, which can be accessed by authorized users. Users can use the same query language to access the catalog which they use to access the database itself.

Rule 5: Comprehensive Data Sub-Language Rule

A database can only be accessed using a language having linear syntax that supports data definition, data manipulation, and transaction management operations. This language can be used directly or by means of some application. If the database allows access to data without any help of this language, then it is considered as a violation.

Rule 6: View Updating Rule

All the views of a database, which can theoretically be updated, must also be updatable by the system.

Rule 7: High-Level Insert, Update, and Delete Rule

A database must support high-level insertion, updation, and deletion. This must not be limited to a single row, that is, it must also support union, intersection and minus operations to yield sets of data records.

Rule 8: Physical Data Independence

The data stored in a database must be independent of the applications that access the database. Any change in the physical structure of a database must not have any impact on how the data is being accessed by external applications.

Rule 9: Logical Data Independence

The logical data in a database must be independent of its user's view (application). Any change in logical data must not affect the applications using it. For example, if two tables are merged or one is split into two different tables, there should be no impact or change on the user application. This is one of the most difficult rule to apply.

Rule 10: Integrity Independence

A database must be independent of the application that uses it. All its integrity constraints can be independently modified without the need of any change in the application. This rule makes a database independent of the front-end application and its interface.

Rule 11: Distribution Independence

The end-user must not be able to see that the data is distributed over various locations. Users should always get the impression that the data is located at one site only. This rule has been regarded as the foundation of distributed database systems.

Rule 12: Non-Subversion Rule

If a system has an interface that provides access to low-level records, then the interface must not be able to subvert the system and bypass security and integrity constraints.

Concepts

Tables − In relational data model, relations are saved in the format of Tables. This format stores the relation among entities. A table has rows and columns, where rows represents records and columns represent the attributes.

Tuple − A single row of a table, which contains a single record for that relation is called a tuple.

Relation instance − A finite set of tuples in the relational database system represents relation instance. Relation instances do not have duplicate tuples.

Relation schema − A relation schema describes the relation name (table name), attributes, and their names.

Relation key − each row has one or more attributes, known as relation key, which can identify the row in the relation (table) uniquely.

Attribute domain − every attribute has some pre-defined value scope, known as attribute domain.

Constraints

Every relation has some conditions that must hold for it to be a valid relation. These conditions are called Relational Integrity Constraints. There are three main integrity constraints −

- Key constraints

- Domain constraints
- Referential integrity constraints

Key Constraints

There must be at least one minimal subset of attributes in the relation, which can identify a tuple uniquely. This minimal subset of attributes is called key for that relation. If there are more than one such minimal subsets, these are called candidate keys.

Key constraints force that −

- In a relation with a key attribute, no two tuples can have identical values for key attributes.
- A key attribute cannot have NULL values.

Key constraints are also referred to as Entity Constraints.

Domain Constraints

Attributes have specific values in real-world scenario. For example, age can only be a positive integer. The same constraints have been tried to employ on the attributes of a relation. Every attribute is bound to have a specific range of values. For example, age cannot be less than zero and telephone numbers cannot contain a digit outside 0-9.

Referential integrity Constraints

Referential integrity constraints work on the concept of Foreign Keys. A foreign key is a key attribute of a relation that can be referred in other relation.

Referential integrity constraint states that if a relation refers to a key attribute of a different or same relation, then that key element must exist.

Data warehouse

A **data warehouse** is a type of data management system that is designed to enable and support business intelligence (BI) activities, especially analytics. Data warehouses are specially intended to perform queries and analysis and often contain large amounts of historical data. The data within a data warehouse is usually derived from a wide range of sources such as application log files and transaction applications. A data warehouse is usually modeled by a multidimensional data structure, called a **data cube**, in which **each dimension corresponds to an attribute or a set of attributes in the schema**, and each cell stores the value of some aggregate measure such as count or sum(sales amount).

Transactional database

A transactional database is a database management system (DBMS) that has the capability to roll back or undo a database transaction or operation if it is not completed appropriately. A relational database systems support transactional database operations.

What is a Database Transaction?

Whenever information or data is stored, manipulated, or "managed" in a database, the operation is considered to be a database transaction. The term indicates that there has been work accomplished within the database management system or in the database itself. In order to ensure security, a database transaction occurs separately from other transactions within the system to ensure that all information stored remains accessible, secure, and coherent.

The idea of conducting transactions with a database rises from the need to provide the end-user or administrator with a means to cohesively and coherently transfer information in a secure manner that is not corrupted by potential system failures. The core properties of database transactions are: atomic, coherent, isolated, and durable, or ACID for short. If a database transaction fails, it will have no impact on the information stored in the database to help ensure these core properties are met.

Some examples of open source and commercial Relational DBMSs

1. **For open source RDBMS, three popular software are MySQL, PostgreSQL, and SQLite.**
2. **For commercial RDBMSs, you can mention Oracle, Microsoft SQL server, IBM DB2, and Teradata.**

Advanced Data and Information Systems and Advanced Applications

1. **Object-Relational Databases**

Object-relational databases are constructed based on an object-relational data model. It is a DBMS where data is represented in the form of objects, as used in object-oriented programming. OODB implements object-oriented concepts such as classes of objects, object identity, polymorphism, encapsulation, and inheritance. An object-oriented database stores complex data as compared to relational database.

2. **Temporal Databases, Sequence Databases, and Time-Series Databases**

Temporal Databases : A temporal database typically stores relational data that include time-related attributes. These attributes may involve several timestamps, each having different semantics. Temporal database is a database that needs some aspect of time for the organization of information. In the temporal database, each tuple in relation is associated with time. It stores information about the states of the real world and time. The temporal database does store information about past states it only stores information about current states. Whenever the state of the database changes, the information in the database gets updated. In many fields, it is very necessary to store information about past states. For example, a stock database must store information about past stock prizes for analysis. Historical information can be stored manually in the schema. A temporal database contains Historical data instead of current data. It provides a uniform way to deal with historical data.

Sequence Databases :A sequence database stores sequences of ordered events, with or without a concrete notion of time. Sequence Data in Data Mining is defined as data in which the points in the dataset are dependent on the other points in the dataset. Examples include customer shopping sequences, Web click streams, and biological sequences.

Time-Series Databases: A time-series database stores sequences of values or events obtained over repeated measurements of time (e.g., hourly, daily, weekly). A **time-series database (TSDB)** is a computer system that is designed to store and retrieve data records that are part of a "time series," which is a set of data points that are associated with timestamps. The timestamps provide a critical context for each of the data points in how they are related to others.

Examples include data collected from the stock exchange, Time series data is often a continuous flow of data like measurements from sensors and intraday stock prices , inventory control, and the observation of natural phenomena (like temperature and wind).

3. Spatial Databases and Spatiotemporal Databases

Spatial databases contain spatial-related information. Examples include geographic (map) databases, very large-scale integration (VLSI) or computed-aided design databases, and medical and satellite image databases. Spatial data may be represented in raster format, consisting of n-dimensional bit maps or pixel maps. For example, a 2-D satellite image may be represented as raster data, where each pixel registers the rainfall in a given area. **Spatiotemporal database** is a database that manages both space and time information. Common examples include:

- Tracking of moving objects, which typically can occupy only a single position at a given time.
- A database of wireless communication networks, which may exist only for a short timespan within a geographic region.

4. **Text Databases and Multimedia Databases :** Text databases are databases that contain word descriptions for objects. These word descriptions are usually not simple keywords but rather long sentences or paragraphs, such as product specifications, error or bug reports, warning messages, summary reports, notes, or other documents. Text databases may be highly unstructured (such as some Web pages on theWorldWideWeb). Some text databases may be somewhat structured, that is, semistructured (such as e-mail messages and many HTML/XML Web pages), whereas others are relatively well structured (such as library catalogue databases). Text databases with highly regular structures typically can be implemented using relational database systems.

Multimedia Databases:

Multimedia databases store image, audio, and video data. They are used in applications such as picture content-based retrieval, voice-mail systems, video-on-demand systems, the World Wide Web, and speech-based user interfaces that recognize spoken commands. Multimedia databases must support large objects.

5. Heterogeneous Databases and Legacy Databases:

A heterogeneous database consists of a set of interconnected, autonomous component databases. The components communicate in order to exchange information and answer queries. Objects in one component database may differ greatly from objects in other component databases, making it difficult to assimilate their semantics into the overall heterogeneous database. Many enterprises acquire legacy databases as a result of the long history of information technology development (including the application of different hardware and operating systems).

Legacy Databases: A legacy database is a group of heterogeneous databases that combines different kinds of data systems, such as relational or object-oriented databases, hierarchical databases, network databases, spreadsheets, multimedia databases, or file systems. Legacy data refers to information that is stored in outdated or obsolete systems, formats, or technologies. This includes structured and unstructured data. It includes data generated by humans as well as machine-generated. While legacy data may not be actively used, at least, if it is used, it's not used regularly, it is often crucial for legal, regulatory, or historical reasons.

6. Data Streams

Many applications involve the generation and analysis of a new kind of data, called stream data, where data flow in and out of an observation platform (or window) dynamically. Data streaming is the continuous transfer of data at a high rate of speed. Many data streams are collecting data from thousands of data sources at the same time. A data stream typically sends large clusters of smaller sized data records simultaneously.

7. The World Wide Web

The World Wide Web (WWW), often called the Web, is a system of interconnected webpages and information that you can access using the Internet. It was created to help people share and find information easily, using links that connect different pages together. The Web allows us to browse websites, watch videos, shop online, and connect with others around the world through our computers and phones.

OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
• Current data	• Historical data
• Day-to-day transactional operations	• Data analysis & decision-making
• Normalized data structures	• Star schema or snowflake schema
• Simple Queries	• Complex Queries
• Used by Front Line Employs, Managers	• Analyst, Executives, Decision-Makers
• Require fast response times	• can have longer response times
• Data in OLTP systems is updated in real time.	• Data in OLAP systems is periodically refreshed
• Oracle, MySQL, SQL Server, DB2	• Tableau, Power BI, and SAP

Types of Data Mining or Data mining functionalities

Each of the Data Mining approaches listed below is useful for a variety of business problems and delivers unique insights into each of them. Understanding the type of business problem you're trying to solve can also help you figure out which strategy to utilize and which will produce the best outcomes. Data Mining classify into two categories : Predictive Data Mining and Descriptive Data Mining

1. **Predictive Data Mining:** It helps developers to provide unlabeled definitions of attributes. Based on previous tests, the software estimates the characteristics that are absent. Predictive mining tasks perform training on the current data in order to make predictions. Predictive Data Mining analysis, as the name suggests, works with data to predict what will happen later (or in the future) in business. Predictive Data Mining is further subdivided into four categories, as indicated below:

- **Classification Analysis:** This data mining technique is commonly used to acquire or retrieve dynamic and relevant information about data and metadata. It can even be used to classify different sorts of data formats into separate groups.
- **Regression Analysis:** Regression analysis is a statistical procedure for determining and analyzing the connection between variables. It means that one variable is dependent on another, but not the other way around. It is typically used for forecasting and prediction.
- **Time Series Analysis:** A time series is a collection of data points that are typically recorded at regular intervals. They are usually – most typically at regular intervals (seconds, hours, days, months etc.).
- **Prediction Analysis:** This method is commonly used to forecast the relationship between the independent and dependent variables, as well as between the independent variables alone. It can also be used to forecast future profit potential based on the transaction.

2. **Descriptive Data Mining:**

It includes certain knowledge to understand what is happening within the data without a previous idea. The common data features are highlighted in the data set. Descriptive mining tasks characterize properties of the data in a target data set. For examples: count, average etc. The primary purpose of Descriptive Data Mining is to summarize or transform given data into useful information. Descriptive Data Mining tasks can alternatively be classified into four categories:

- **Clustering Analysis:** This technique is used in Data Mining to construct meaningful item clusters with similar features. Clustering puts objects in classes that are determined by it, unlike classification, which collects objects into predefined classes.
- **Summarization Analysis:** Summarization analysis is used to store a group (or set) of data in a more compact and understandable format.
- **Association Rules Analysis:** It can be thought of as a strategy for identifying some interesting relationships (dependency modeling) between distinct variables in huge databases in general. This technique can also assist us in uncovering certain hidden patterns in the data that can be utilized to identify variables.
- **Sequence Discovery Analysis:** Sequence discovery analysis' main purpose is to find interesting patterns in data based on some subjective or objective evaluation of how interesting it is. This assignment usually entails finding frequent sequential patterns in relation to a frequency support measure.

Classification of Data Mining Systems

- **Classification according to kinds of techniques used** : We can classify the data mining system according to kind of techniques used. We can describes these techniques according to degree of user interaction involved or the methods of analysis employed.
- **Classification according to applications adapted**: We can classify the data mining system according to application adapted. These applications are as follows: Finance, Telecommunications, DNA, Stock Markets, E-mail

- **Classification according to the types of knowledge mined:**
  This is based on functionalities such as characterization, association, discrimination and correlation, prediction etc.
- **Classification according to types of databases mined:**
  A database system can be classified as a ‘type of data’ or ‘use of data’ model or ‘application of data’.

Data Preprocessing ______Data Cleaning: Data Cleaning Real-world data tend to be incomplete, noisy, and inconsistent. Data cleaning (or data cleansing) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data. The basic methods for data cleaning are-

Missing Values

1. Ignore the tuple
2. Fill in the missing value manually
3. Use a global constant to fill in the missing value: Replace all missing attribute values by the same constant, such as a label like "Unknown" or $-\infty$. If missing values are replaced by, say, "Unknown," then the mining program may mistakenly think that they form an interesting concept
4. Use the attribute mean to fill in the missing value:
5. Use the most probable value to fill in the missing value

1. **Noisy Data** : Noise is a random error or variance in a measured variable. Let's look at the following data smoothing techniques:

   Binning: Binning methods smooth a sorted data value by consulting its "neighborhood," that is, the values around it. The sorted values are distributed into a number of "buckets," or bins. Because binning methods consult the neighborhood of values, they perform local smoothing

   Sorted data for price (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34 Partition into (equal-frequency)

   Bin: Bin 1: 4, 8, 15 Bin 2: 21, 21, 24 Bin 3: 25, 28, 34

   Smoothing by bin means: Bin 1: 9, 9, 9 Bin 2: 22, 22, 22 Bin 3: 29, 29, 29

   Smoothing by bin boundaries: Bin 1: 4, 4, 15 Bin 2: 21, 21, 24 Bin 3: 25, 25, 34

   Binning methods for data smoothing. 1. Binning: Binning methods smooth a sorted data value by consulting its "neighborhood," that is, the values around it. The sorted values are distributed into a number of "buckets," or bins. Because binning methods consult the neighborhood of values

Regression: Data can be smoothed by fitting the data to a function, such as with regression. Linear regression involves finding the "best" line to fit two attributes (or variables), so that one attribute can be used to predict the other. Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

Clustering: Outliers may be detected by clustering, where similar values are organized into groups, or "clusters." Intuitively, values that fall outside of the set of clusters may be considered outliers

Data Transformation

In data transformation, the data are transformed or consolidated forms which is appropriate for mining. Data transformation can involve the following:

Aggregation, where summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts.

Generalization of the data, where low-level or "primitive" (raw) data are replaced by higher-level concepts through the use of concept hierarchies. Attributes, like street, can be generalized to higher-level concepts, like city or country. Similarly, values for numerical attributes, like age, may be mapped

youth, middle-aged, and senior.

Normalization, where the attribute data are scaled so as to fall within a small specified range, such as −1.0 to 1.0, or 0.0 to 1.0.

$$v' = \frac{v - \bar{A}}{\sigma_A}, \tag{2.12}$$

where $\bar{A}$ and σ_A are the mean and standard deviation, respectively, of attribute A. This method of normalization is useful when the actual minimum and maximum of attribute A are unknown, or when there are outliers that dominate the min-max normalization.

Example 2.3 **z-score normalization** Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for *income* is transformed to $\frac{73,600-54,000}{16,000} = 1.225$. ■

Normalization by decimal scaling normalizes by moving the decimal point of values of attribute A. The number of decimal points moved depends on the maximum absolute value of A. A value, v, of A is normalized to v' by computing

$$v' = \frac{v}{10^j}, \tag{2.13}$$

where j is the smallest integer such that $Max(|v'|) < 1$.

Example 2.4 **Decimal scaling.** Suppose that the recorded values of A range from −986 to 917. The maximum absolute value of A is 986. To normalize by decimal scaling, we therefore divide each value by 1,000 (i.e., $j = 3$) so that −986 normalizes to −0.986 and 917 normalizes to 0.917. ■

Data Reduction in Data Mining: Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information or technique used to reduce the number of features. Transforming high-dimensional data into a lower-dimensional space that still preserves the essence of the original data. This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.

There are several different data reduction techniques that can be used in data mining, including:

1. **Data Sampling:** This technique involves selecting a subset of the data to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.
2. **Dimensionality Reduction:** This technique involves reducing the number of features in the dataset, either by removing features that are not relevant or by combining multiple features into a single feature.
3. **Data Compression:** This technique involves using techniques such as lossy or lossless compression to reduce the size of a dataset.
4. **Data Discretization:** This technique involves converting continuous data into discrete data by partitioning the range of possible values into intervals or bins.
5. **Feature Selection:** This technique involves selecting a subset of features from the dataset that are most relevant to the task at hand.

In conclusion, data reduction is an important step in data mining, as it can help to improve the efficiency and performance of machine learning algorithms by reducing the size of the dataset. However, it is important to be aware of the trade-off between the size and accuracy of the data, and carefully assess the risks and benefits before implementing it.

Methods of data reduction:

These are explained as following below.

1. Data Cube Aggregation:

This technique is used to aggregate data in a simpler form. For example, imagine the information you gathered for your analysis for the years 2012 to 2014, that data includes the revenue of your company every three months. They involve you in the annual sales, rather than the quarterly average, So we can summarize the data in such a way that the resulting data summarizes the total sales per year instead of per quarter. It summarizes the data.

2. Dimension reduction: There are two main approaches to dimensionality reduction: feature selection and feature extraction.

a. Feature Selection:

Feature selection involves selecting a subset of the original features that are most relevant to the problem at hand. The goal is to reduce the dimensionality of the dataset while retaining the most important features. There are several methods for feature selection, including **filter methods, wrapper methods**, and **embedded methods**. Filter methods rank the features based on their relevance to the target variable, wrapper methods use the model performance as the criteria for selecting features, and embedded methods combine feature selection with the model training process.

b. Feature Extraction:

Feature extraction involves creating new features by combining or transforming the original features. The goal is to create a set of features that captures the essence of the original data in a lower-dimensional space. There are several methods for feature extraction, including principal component analysis (PCA), linear discriminant analysis (LDA) etc. PCA is a popular technique that projects the original features onto a lower-dimensional space while preserving as much of the variance as possible.

Whenever we come across any data which is weakly important, then we use the attribute required for our analysis. It reduces data size as it eliminates outdated or redundant features.

- **Step-wise Forward Selection –**
  The selection begins with **an empty set** of attributes later on we decide the **best of the original attributes** on the set based on their relevance to other attributes. We know it as a p-value in statistics.

  Suppose there are the following attributes in the data set in which few attributes are redundant.

```
Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set:  { }

Step-1: {X1}
Step-2: {X1, X2}
Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}
```

- **Step-wise Backward Selection –**
  This selection starts with **a set of complete attributes** in the original data and at each point, it **eliminates the worst remaining attribute** in the set.

  Suppose there are the following attributes in the data set in which few attributes are redundant.

```
Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set:  {X1, X2, X3, X4, X5, X6 }

Step-1: {X1, X2, X3, X4, X5}
Step-2: {X1, X2, X3, X5}
Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}
```

- **Combination of forwarding and Backward Selection –**
  It allows us to remove the worst and select the best attributes, saving time and making the process faster.

3. Data Compression:
The data compression technique reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

- **Lossless Compression –**
  Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms **to restore the precise original data from** the compressed data. Example: text data
- **Lossy Compression –**
  Methods such as the Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. For e.g., the **JPEG image format is a lossy compression**, but we can find the meaning equivalent to the original image. In lossy-data compression, the **decompressed data may differ from the original data** but are useful enough to retrieve information from them.

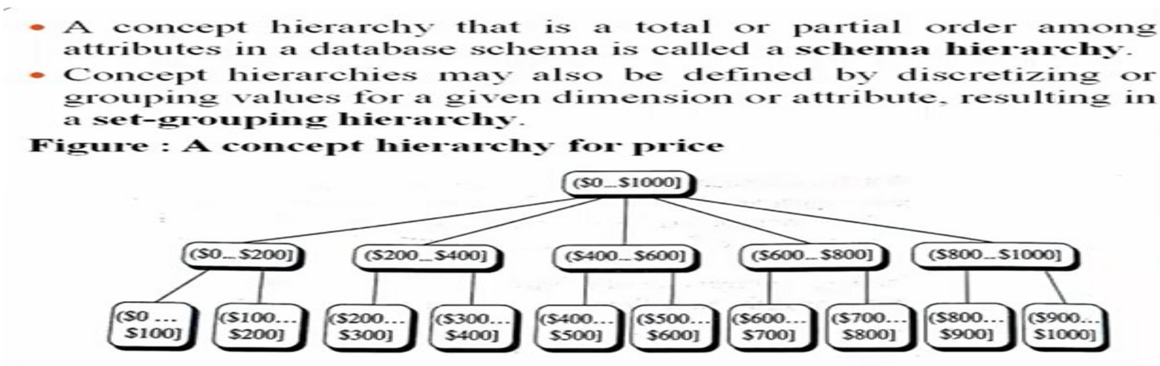
4. Numerosity Reduction:
In this reduction technique, the actual data is replaced with mathematical models or **smaller representations of the data instead of actual data**, it is important to only store the model **parameter. Parametric** methods such as regression and **non-parametric** methods such as clustering, histogram, and sampling.

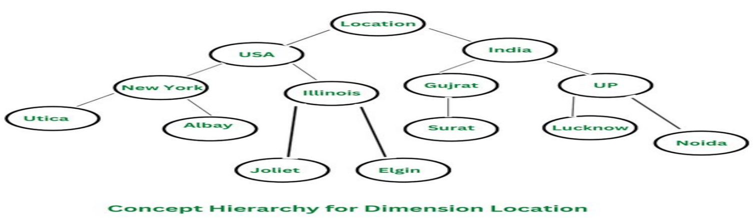
5. Discretization & Concept Hierarchy Operation:
Techniques of data discretization are used to **divide the attributes of the continuous nature into data with intervals**. We replace many constant values of the attributes by labels of **small intervals**. This means that mining results are shown in a brief, and easily understandable way.

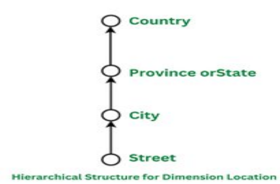
- **Top-down discretization –** (splitting)
  If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat this method up to the end, then the process is known as top-down discretization also known as splitting.
- **Bottom-up discretization –** (merging)
  If you first consider all the constant values as split points, some are discarded through a combination of the neighborhood values in the interval, that process is called bottom-up discretization.

Concept Hierarchies: Concept Hierarchy in Data Mining

The concept hierarchy define as a sequence of mappings as a low level concepts to high level concepts. In data mining, the concept of a concept hierarchy refers to the organization of data into a tree-like structure, where each level of the hierarchy represents a concept. This hierarchical organization of data allows for more efficient and effective data analysis, as well as the ability to drill down to more specific levels of detail when needed. The concept of hierarchy is used to organize and classify data in a way that makes it more understandable and easier to analyze. The main idea behind the concept of hierarchy is that the same data can have different levels of granularity or levels of detail and that by organizing the data in a hierarchical fashion, it is easier to understand and perform analysis.







Role and Importance Concept Hierarchy in Data Mining

Concept hierarchies play a pivotal role in various data mining tasks, such as data discretization, data summarization, and association rule mining. They help in:

- **Data Discretization:** Transforming continuous data into categorical data by dividing it into intervals or clusters. Concept hierarchies facilitate this by defining how data should be grouped.
- **Data Summarization:** Providing a compact representation of data by aggregating detailed data into higher-level summaries. This is particularly useful for generating reports and dashboards.
- **Association Rule Mining:** Identifying interesting correlations and relationships between data items. Concept hierarchies help in generalizing rules to make them more meaningful and interpretable.

Data Objects and Attribute Types

Data sets are made up of data objects. **A data object represents an entity**—in a sales database, the objects may be customers, store items, and sales; in a medical database, the objects may be patients; in a university database, the objects may be students, professors, and courses. **Data objects are typically described by attributes.** Data objects can also be referred to as samples, examples, instances, data points, or objects. If the data objects are stored in a database, they are data tuples. That is, the **rows of a database correspond to the data objects**, and the **columns correspond to the attributes.**

What Is an Attribute?

An attribute is a data field, representing a characteristic or feature of a data object. The nouns, attribute, dimension, feature, and variable are often used interchangeably in the literature. **The term dimension is commonly used in data warehousing**.

Machine learning literature tends to use the **term feature**, while **statisticians prefer** the term **variable**. **Data mining and database professionals** commonly use the term **attribute**, and we do here as well. Attributes describing a customer object can include, for example, customer ID, name, and address. **Observed values** for a given attribute are known as **observations**. A **set of attributes** used to describe a given object is called an **attribute vector** (or feature vector). The distribution of data involving **one attribute** (or variable) is called u**nivariate**. A **bivariate** distribution **involves two attributes**, and so on.

The type of an attribute is determined by the set of possible values—nominal, binary, ordinal, or numeric—the attribute can have. In the following subsections, we introduce each type.

1. **Nominal Attributes**

Nominal means "relating to names." The values of a nominal attribute are **symbols or names of things**. Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as **categorical**. The values do not have any meaningful order. In computer science, the values are also known as details.

Example 1. Nominal attributes. Suppose that hair color is an attributes describing person objects. In our application, possible values for hair color are black, brown, blond, red, and white.

Another example of a **nominal attribute is occupation,** with the values **teacher, dentist, programmer, farmer, and so on.**

Although we said that the values of a nominal attribute are symbols or "names of things," it is possible to represent such symbols or "names" with numbers. With hair color, for instance, we can assign a code of 0 for black, 1 for brown, and so on.

2. **Binary Attributes**

A binary attribute is a nominal attribute with only two categories or states: 0 or 1, where 0 typically means that the attribute is absent, and 1 means that it is present. Binary attributes are referred to as Boolean if the two states correspond to true and false.

Example 2.2 Binary attributes. Given the attribute smoker describing a patient object, 1 indicates that the patient smokes, while 0 indicates that the patient does not. Similarly, suppose the patient undergoes a medical test that has two possible outcomes. The attribute medical test is binary, where a value of 1 means the result of the test for the patient is positive, while 0 means the result is negative.

Binary attribute are two types

a. **Symmetric binary attribute**

A binary attribute is symmetric if both of its states are equally valuable and carry the same weight; that is, there is no preference on which outcome should be coded as 0 or 1. One such **example could be the attribute gender** having the states male and female.

b. **Asymmetric binary attribute**

A binary attribute is **asymmetric** if the outcomes of the states are not equally important, such as the positive and negative outcomes of a medical test for HIV. By convention, we code the most important outcome, which is usually by 1 (e.g., HIV positive) and the other by 0 (e.g., HIV negative).

3. **Ordinal Attributes**

An ordinal attribute is an attribute with possible values that have **a meaningful order or ranking among them, but the magnitude between successive values is not known.**

Example : Ordinal attributes. Suppose that drink size corresponds to the size of drinks available at a fast-food restaurant. This nominal attribute has three possible values: small, medium, and large. The values have a meaningful sequence (which corresponds to increasing drink size); however, we cannot tell from the values how much bigger, say, a medium is than a large. Other examples of **ordinal attributes include grade** (e.g., A+,A,B and so on) and professional rank. Professional ranks can be listed in a sequential order: for example, assistant, associate, and full for professors, and private, private first class, specialist, corporal, and sergeant for army ranks

4. Numeric Attributes

A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values. Numeric attributes can be **interval-scaled or ratio-scale**

a. Interval-Scaled Attributes

Interval-scaled attributes are measured on a scale of equal-size units. The values of interval-scaled attributes have order and can be positive, 0, or negative. Thus, in addition to providing a ranking of values, such attributes allow us to compare and calculate the difference between values.

Example 2.4 Interval-scaled attributes**. A temperature attribute is interval-scaled**. Suppose that we have the outdoor temperature value for a number of different days, where each day is an object. By ordering the values, we obtain a ranking of the objects with respect to temperature.

b. Ratio-Scaled Attributes

3. **A ratio-scaled attribute is a numeric attribute with an inherent zero-point**. That is, if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value. Ratio scale provides unique possibilities for statistical analysis. In this scale, variables can be systematically added, subtracted, multiplied and divided (ratio). All statistical analysis including mean, mode, and median can be calculated using it.

Example 2.5 Ratio-scaled attributes. Unlike temperatures in Celsius and Fahrenheit, the Kelvin (K) temperature scale has what is considered a true zero-point

c. Discrete versus Continuous Attributes

Classification algorithms developed from the field of machine learning often talk of attributes as being either discrete or continuous. Each type may be processed differently.

A discrete attribute has a finite or count ably infinite set of values, which may or may not be represented as integers. The attributes hair color, smoker, medical test, and drink size each have a finite number of values, and so are discrete. Note that discrete attributes may have numeric values, such as 0 and 1 for binary attributes or, the values 0 to 110 for the attribute age. **An attribute is countably infinite if the set of possible values is infinite but the values can be put in a one-to-one correspondence with natural numbers**. For example, the attribute customer ID is countably infinite. The number of customers can grow to infinity, but in reality, the actual set of values is countable (where the values can be put in one-to-one correspondence with the set of integers). Zip codes are another
example.

If an attribute is not discrete, it is continuous. The terms numeric attribute and continuous attribute are often used interchangeably in the literature. (This can be confusing because, in the classic sense, continuous values are real numbers, whereas numeric values can be either integers or real numbers.) In practice, real values are represented using a finite number of digits. Continuous attributes are typically represented as floating-point variables.

Issues of Data Mining

Data mining systems face a lot of challenges and issues in today's world some of them are:

2. **Mining methodology and user interaction issues:**

a. **Mining different kinds of knowledge in databases:**

Different user - different knowledge - different way. That means different client want a different kind of information so it becomes difficult to cover vast range of data that can meet the client requirement.

b. **Interactive mining of knowledge at multiple levels of abstraction:**

Interactive mining allows users to focus the search for patterns from different angles. The data mining process should be interactive because it is difficult to know what can be discovered within a database.

c. **Incorporation of background knowledge:**

Background knowledge is used to guide discovery process and to express the discovered patterns.

d. **Query languages and ad hoc mining:**

Relational query languages (such as SQL) allow users queries for data retrieval. The language of data mining query language should be in perfectly matched with the query language of data warehouse.

e. **Handling noisy or incomplete data:**

In a large database, many of the attribute values will be incorrect. This may be due to human error or because of any instruments fail. Data cleaning methods and data analysis methods are used to handle noise data.

2 Performance issues

b. **Efficiency and scalability of data mining algorithms:**

To effectively extract information from a huge amount of data in databases, data mining algorithms must be efficient and scalable.

c. **Parallel, distributed, and incremental mining algorithms:**

The huge size of many databases, the wide distribution of data, and complexity of some data mining methods are factors motivating the development of parallel and distributed data mining algorithms. Such algorithms divide the data into partitions, which are processed in parallel.

3 Issues relating to the diversity of database types:

Handling of relational and complex types of data:

There are many kinds of data stored in databases and data warehouses. It is not possible for one system to mine all these kind of data. So different data mining system should be construed for different kinds data.

Mining information from heterogeneous databases and global information systems:

Since data is fetched from different data sources on Local Area Network (LAN) and Wide Area Network (WAN).The discovery of knowledge from different sources of structured is a great challenge to data mining.